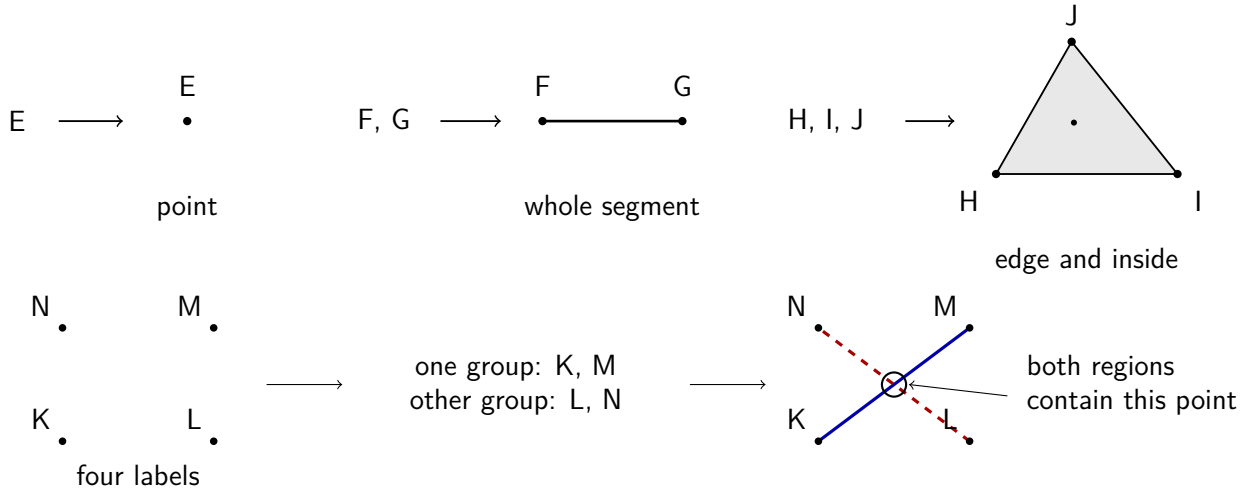
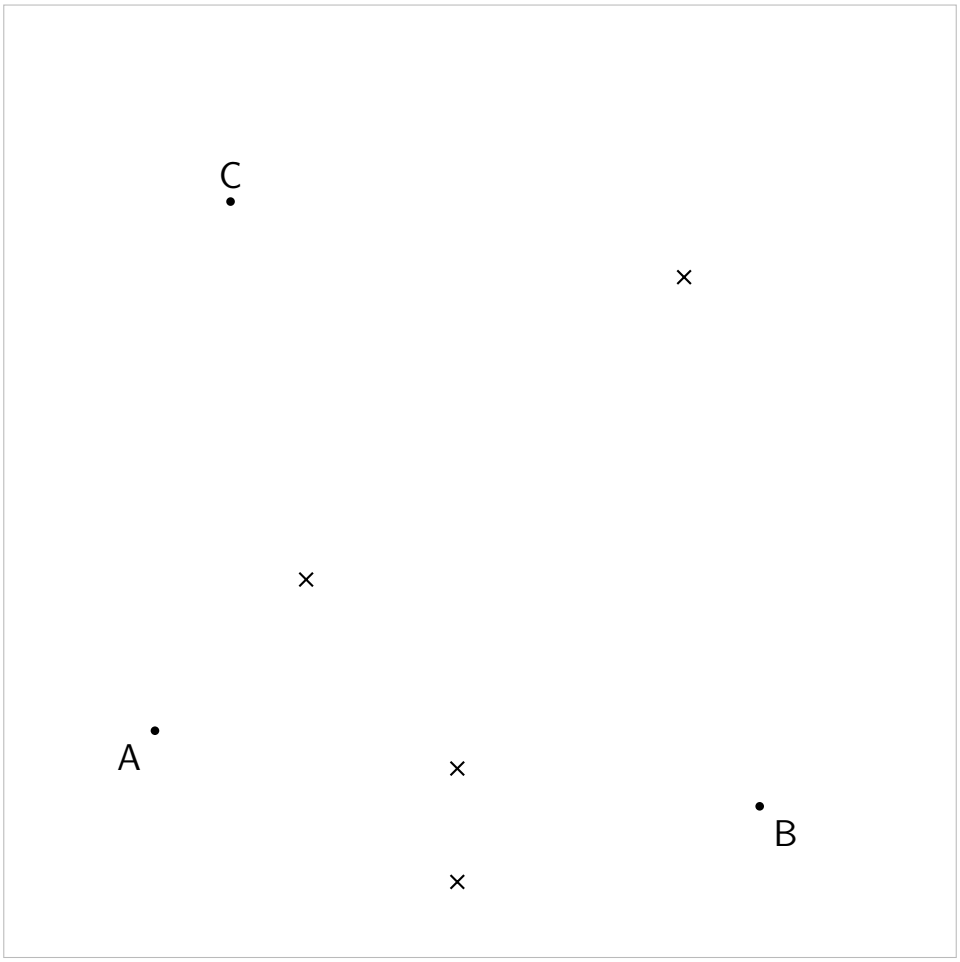
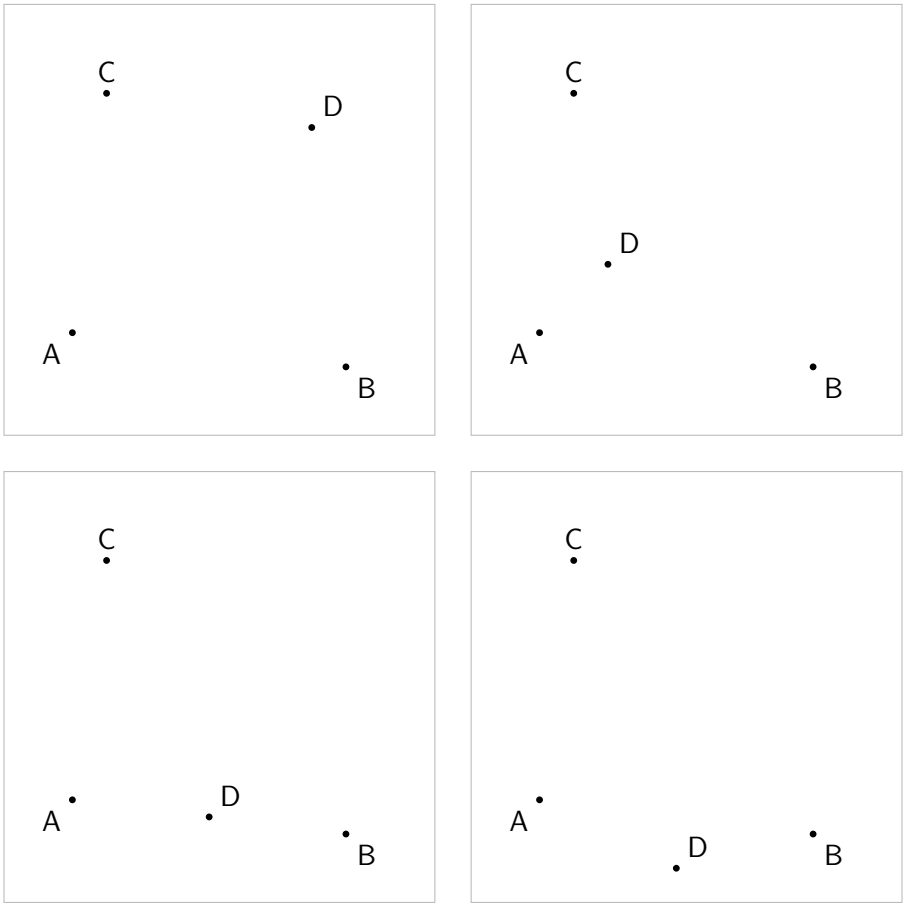
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split. When dots lie on one line, draw one group's region just above them and the other group's region just below.



Problem 1: Move the D marker to each cross in turn. Each time, use the markers and your finger to find a split of A, B, C, and D whose two regions meet.

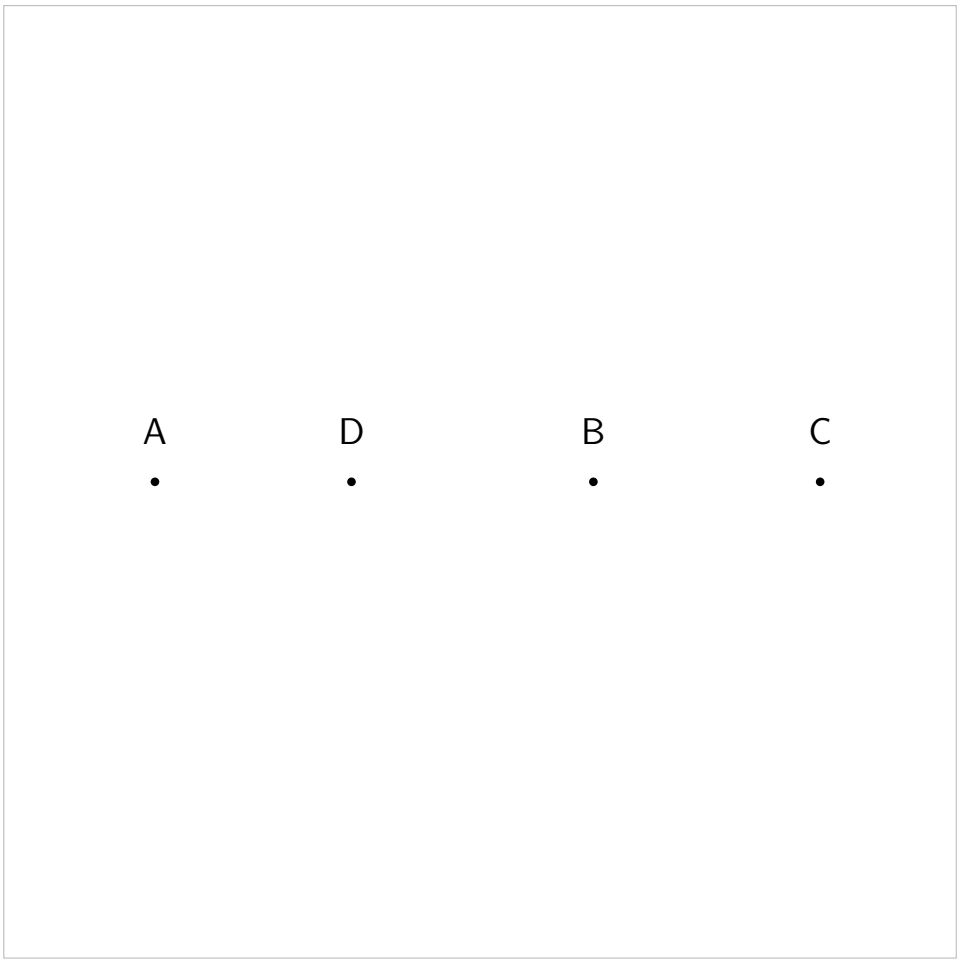


Problem 1: Draw the split you found for each position of D, and mark a point its two regions share.

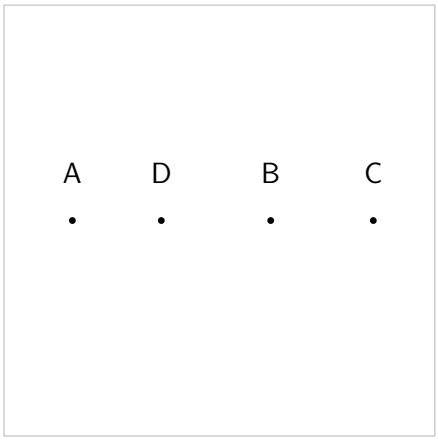


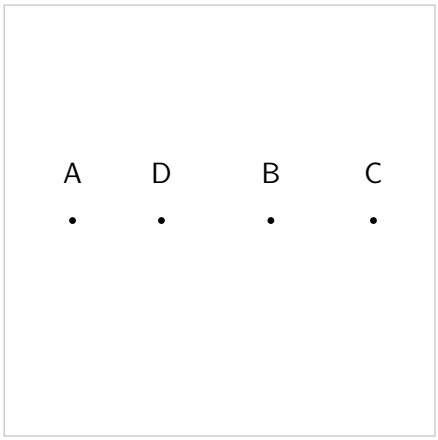
Problem 2: Take turns placing A, B, C, and D in the box. Try to make an arrangement that the other player cannot split successfully. Can four dots always be split so their regions meet?

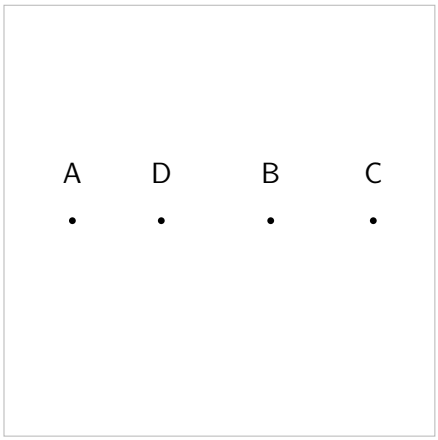
Problem 3: Find every split of A, B, C, and D whose regions meet. Explain how you know your list is complete.

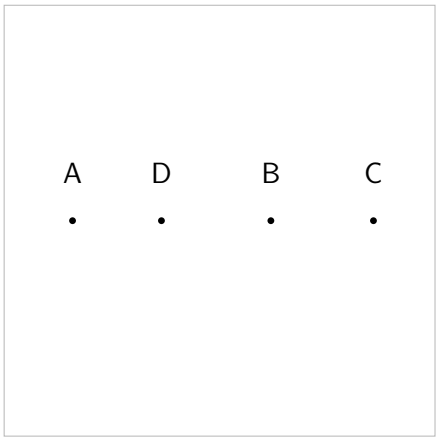


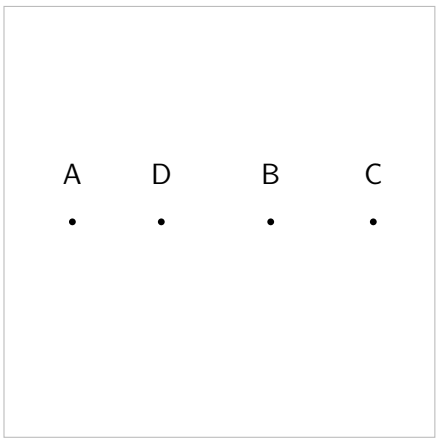
Problem 3: Find every split of A, B, C, and D whose regions meet. Explain how you know your list is complete.

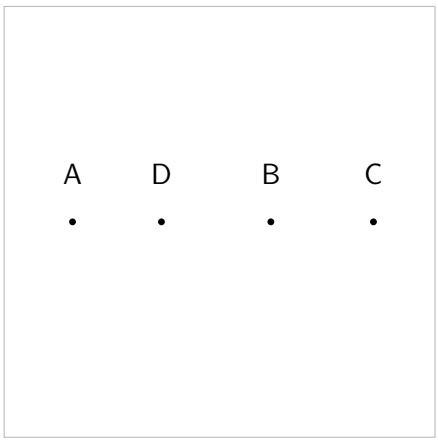










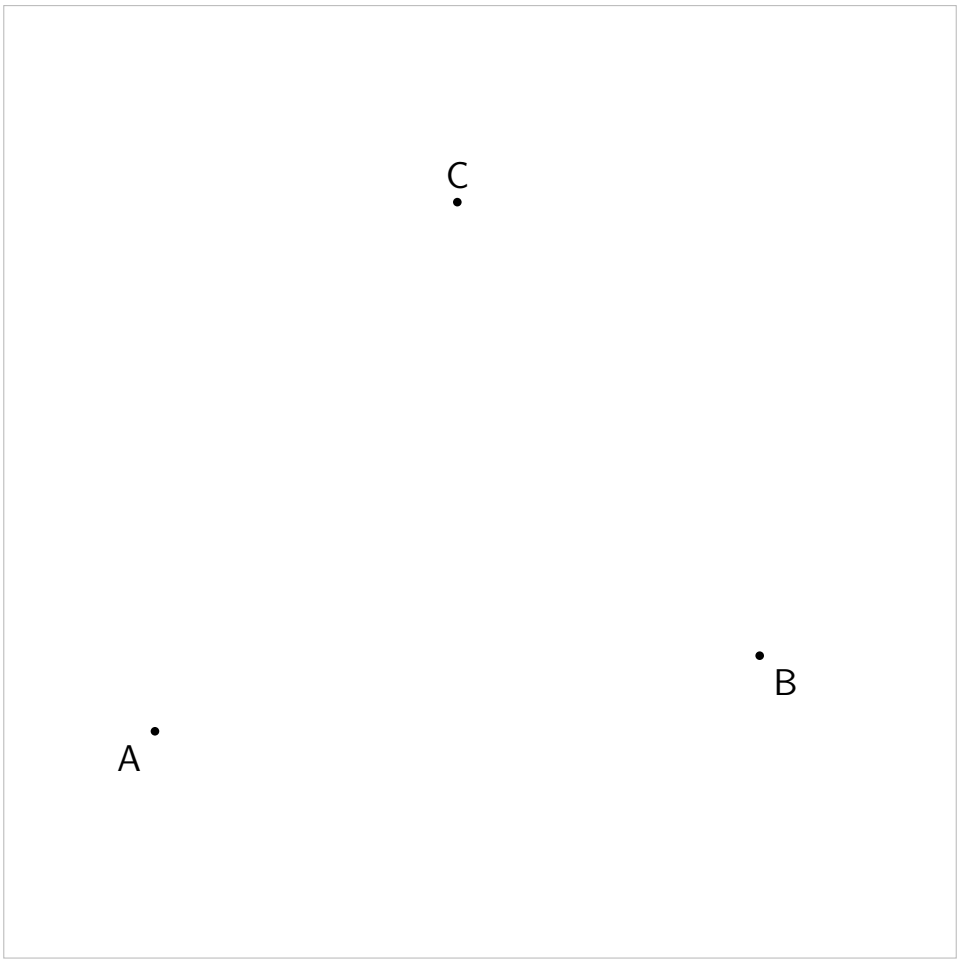


Problem 4: Place A, B, C, and D so that their two regions share a whole straight segment. Can two groups made from these four labels share a filled triangle? Explain.

Problem 5: Make two arrangements of A, B, C, and D that have no working split in common. Show a working split for each.

Problem 5: Make two arrangements of A, B, C, and D that have no working split in common. Show a working split for each.

Problem 6: Can these three dots be split into two groups whose regions meet? Explain why your answer covers every split.



Problem 7: Keep A and B where they are, and put C on each cross in turn. Circle each cross where A, B, and C can be split into two groups whose regions meet.

